CHAPTER 17

PLA, PAL & PLD (Programmable Logic)

Programmable arrays · ROM as logic · PLA · PAL · GAL, CPLD, FPGA · fuse counting

QUICK TRICK

One line to learn first: PROM = programmable OR, PAL = programmable AND (the A in PAL), PLA = both programmable. Every function is an **AND plane** (making product terms) followed by an **OR plane** (adding them).

17.1 What is programmable logic?

● **LEVEL 1 · BEGINNER**

The problem. Making a special chip for every new circuit is slow and expensive. **The idea.** Make **one standard chip** that contains a grid of wires and gates, with **programmable connections**. By choosing which connections to keep, you build **your own** logic function inside the chip.

Analogy. A pegboard or a blank switchboard. All the wires are there. You decide which ones to connect.

Why does this work? Any Boolean function can be written as **SOP** (Chapter 6): AND gates make the product terms, an OR gate adds them. So a chip with an **AND plane** followed by an **OR plane** can do any function.

Word	Simple meaning
PLD	**P**rogrammable **L**ogic **D**evice (the general name).
Product term	One AND term, like AB' (a row in the AND plane).
AND plane / array	Makes the product terms from the inputs and their complements.
OR plane / array	Adds chosen product terms to make each output.
Fuse / programmable connection	A link that is kept (1) or cut (0) when the chip is programmed.

EASY MEMORY

In the drawings: a **filled dot** = a programmable connection that you chose to keep. A **hollow square** = a connection that is **fixed** by the maker.

17.2 ROM as a programmable logic device (PROM)

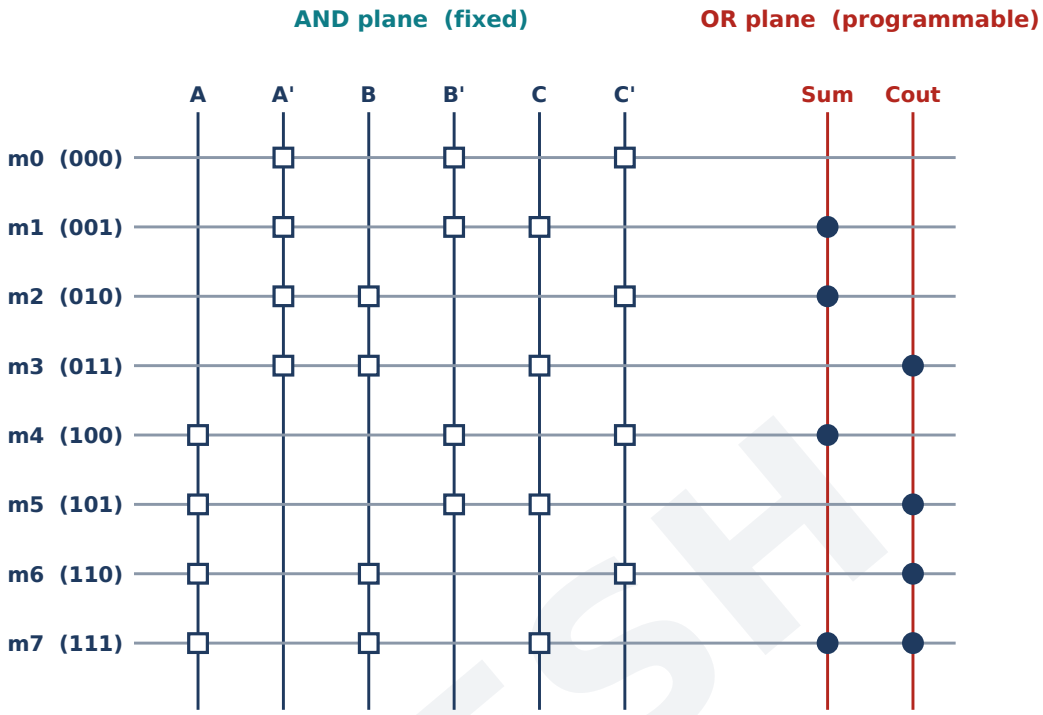
● **LEVEL 2 · EXAM**

A **PROM** has a **fixed AND plane** (a decoder that makes **all 2^n minterms**) and a **programmable OR plane**. So any output is the OR of the minterms you choose. **The inputs act as the address, the outputs are the stored data.**

QUICK TRICK

ROM size = 2^n words × m bits for n inputs and m outputs. The full adder: $2^3 \times 2 =$ **16** programmable bits. **Problem:** the ROM always makes **all 2^n** minterms, even the ones you do not need. A 16-input function would need 65,536 product lines! That is why PLA and PAL were invented: they make only the product terms you need.

Example: a full adder in a ROM (3 inputs A, B, Cin and 2 outputs Sum, Cout). The ROM has 8 words of 2 bits.



Every row is a minterm (made by the fixed AND plane). We keep a dot where the minterm belongs to the output: Sum = Σm(1, 2, 4, 7) and Cout = Σm(3, 5, 6, 7).

17.3 PLA (Programmable Logic Array)

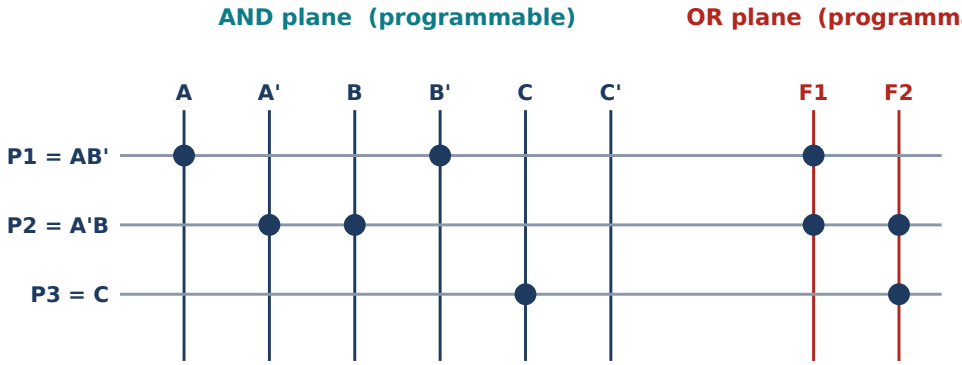
● LEVEL 2 · EXAM

What is it? A PLA has **both planes programmable**. You choose exactly which product terms to make, and each output chooses which product terms to add. A product term can be **shared** by many outputs.

Example. F1 = AB' + A'B (XOR) and F2 = A'B + C. The term **A'B is used by both outputs**, so the PLA needs only **3 product terms**.

Product term	A	B	C	F1	F2
P1 = AB'	1	0	–	✔	
P2 = A'B	0	1	–	✔	✔
P3 = C	–	–	1		✔

(In the AND part: 1 = use the input, 0 = use its complement, – = input not used. In the OR part: ✔ = add this term to the output.)



QUICK TRICK

How to read the drawing: go along a row (product line). The dots in the AND plane tell you which inputs are ANDed. The dots in the OR plane tell you which outputs use that row. **Row P2** has two OR dots, so A'B goes to both F1 and F2.

Counting the programmable connections (fuses)

```
PLA with n inputs, k product terms, m outputs
     AND plane :  2 x n x k  crosspoints      (each input and its complement)
     OR plane  :      k x m  crosspoints
     Total     :  2nk + km
```

Example (Level 2): the PLA above

n = 3, k = 3, m = 2. AND plane: 2 × 3 × 3 = 18. OR plane: 3 × 2 = 6. **Total = 24** programmable points.

Example (Level 3): a PLA of size 16 × 48 × 8

Given: n = 16 inputs, k = 48 product terms, m = 8 outputs.

Concept: total = 2nk + km.

Calculation: 2 × 16 × 48 = 1536. 48 × 8 = 384. 1536 + 384 = 1920.

Answer: 1920 programmable connections.

Example (Level 3): full adder in a PLA vs a ROM

Sum has 4 product terms (A'B'C, A'BC', AB'C', ABC), Cout has 3 (AB, BC, AC). Nothing is shared → k = 7.

PLA: 2 × 3 × 7 + 7 × 2 = **56** points. ROM: only **16** bits of OR plane (but its AND plane has 8 fixed rows).

For a function with **few** product terms but **many inputs**, PLA wins hugely. For a function with **many** minterms and few inputs, a ROM can be just as good.

QUICK TRICK

To size a PLA: 1) simplify every output function (K-map). 2) Collect all the **different** product terms (count shared terms **once**). 3) That is k. Then use 2nk + km.

GATE POINT

Some PLAs also have an **XOR gate on each output** with a fuse. This lets you choose to produce F or F' (whichever needs fewer product terms).

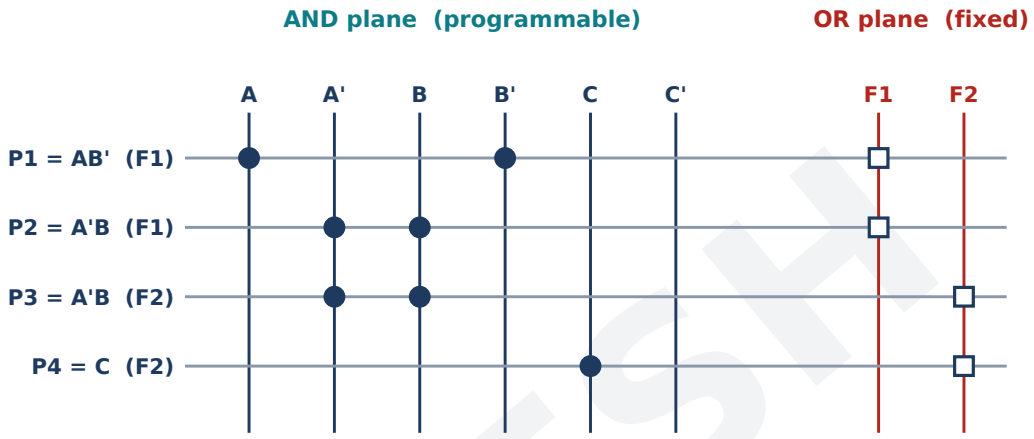
17.4 PAL (Programmable Array Logic)

● LEVEL 2 · EXAM

What is it? A PAL has a **programmable AND plane** but a **fixed OR plane**. Each output is connected to its **own fixed group** of product terms. **Why?** Only one plane is programmable, so it is **cheaper, smaller and faster**.

The price: a product term **cannot be shared**. If two outputs need A'B, the PAL must make A'B twice. And each output is limited to the number of product terms it has.

The same two functions on a PAL (each output has 2 fixed product terms):



QUICK TRICK

Notice: A'B appears **twice** (P2 and P3). In the PLA it was shared. In the PAL each output has its own terms. **PAL fuses = 2 × n × (total product terms of all outputs)**. Here: 2 × 3 × 4 = 24 (only the AND plane is programmable).

EASY MEMORY

PAL families: names like PAL16L8 or PAL16R4 give the number of inputs (16), the type of output (**L** = active-low logic output, **R** = **R**egistered, that is, with a flip-flop) and the number of outputs (8 or 4). **GAL** (Generic Array Logic) is a PAL that can be erased and reprogrammed.

COMMON MISTAKE

A PAL is NOT able to share product terms and its OR plane is NOT programmable. Do not mix it up with the PLA.

17.5 PROM vs PAL vs PLA

● LEVEL 2 · EXAM

	PROM	PAL	PLA
AND plane	**Fixed** (all minterms)	**Programmable**	**Programmable**
OR plane	**Programmable**	**Fixed**	**Programmable**
Product terms	All 2^n minterms	Limited per output	Chosen freely (shared)
Sharing a term between outputs	Yes (outputs pick any minterm)	**No**	**Yes**
Flexibility	Low for big n	Medium	**Highest**
Cost / speed	Large for many inputs	**Cheapest, fastest**	More costly, slower

	PROM	PAL	PLA
Best for	Small n, lookup tables, code	Common simple logic	Complex logic with shared terms

QUICK TRICK
Memory line: PROM = fixed AND, programmable OR. **PAL** = programmable AND, fixed OR. **PLA** = both programmable (most flexible). Fewer programmable planes = cheaper and faster.

17.6 The PLD family: SPLD, CPLD, FPGA

● **LEVEL 1 · BEGINNER**

Device	What it is	Typical size
SPLD (simple PLD)	PROM, PLA, PAL, GAL: one AND–OR block	A few hundred gates
CPLD (complex PLD)	Many PAL-like blocks joined by a programmable switch matrix. Non-volatile (keeps its program when the power is off).	Thousands of gates
FPGA (field-programmable gate array)	A sea of small blocks (**lookup tables** + flip-flops) joined by programmable wires. Usually SRAM-based, so it is **volatile**: it must be reloaded at power-up.	Millions of gates

Lookup table (LUT): a tiny memory that acts as logic

An **n-input LUT** stores $\mathbf{2^n}$ **bits** (the truth table of any n-input function) and a 2^n:1 MUX picks the right bit. A 4-input LUT stores **16 bits** and can be any function of 4 variables. This is the same idea as the ROM and the MUX from Chapter 10.

Programming technology	Can be erased?	Keeps program without power?	Used in
Fuse (blow the link)	No (one time)	Yes	PROM, PAL
Anti-fuse (join the link)	No (one time)	Yes	Some FPGAs
EPROM / EEPROM / Flash cell	Yes	Yes	GAL, CPLD
SRAM cell	Yes (reloaded)	**No** (volatile)	Most FPGAs

Quick-Solve Tricks — Summary

Topic	Trick
Which plane is programmable?	PROM: OR. PAL: AND (A in PAL). PLA: both.
ROM size	$2^n \times m$ bits.
PLA crosspoints	$2nk + km$ (n inputs, k product terms, m outputs).
PAL programmable points	$2 \times n \times$ (total product terms).
Counting k	Simplify first. Count each different product term once for a PLA (shared terms count once).
Sharing terms	PLA yes, PAL no.

Topic	Trick
Cheaper and faster	PAL (only one programmable plane).
LUT	n-input LUT = 2^n bits.
Volatile PLD	SRAM-based FPGA.

GATE QUICK CHECK — Chapter 17

```
ROM : 2^n x m bits        PLA : 2nk + km programmable points
PAL : AND plane programmable, OR plane fixed     PLA : both programmable
n-input LUT = 2^n bits    SRAM FPGA = volatile
```

Traps: PAL vs PLA (which plane is fixed) · shared product terms need a PLA · count k after simplifying · each input needs two lines (true and complement) in the AND plane · ROM makes all minterms.

Practice questions

● **LEVEL 1 · BEGINNER**

Q1. Which plane of a PAL is fixed?

Q2. What does PLD stand for?

Q3. Which device has both the AND plane and the OR plane programmable?

● **LEVEL 2 · EXAM**

Q4. A PLA has 4 inputs, 6 product terms and 3 outputs. How many programmable crosspoints does it have?

Q5. How many bits does a ROM with 3 inputs and 2 outputs have?

Q6. How many product lines (rows) must the fixed AND plane of a 4-input ROM have?

● **LEVEL 3 · GATE**

Q7. A PLA has 16 inputs, 48 product terms and 8 outputs. How many programmable connections?

Q8. F1 = AB' + A'B and F2 = A'B + C. How many different product terms does a PLA need? And how many rows does a PAL (each output separate) need?

Q9. How many bits does a 4-input LUT store? What does it contain?

Q10. Which of PROM, PAL and PLA can share one product term between two outputs?

Q11. A 6-input, 1-output function is stored in a ROM. How many words does the ROM have?

Q12. Which type of programmable technology loses its program when the power is switched off? Give an example device.

Answers

Q	Solution
1	The **OR plane** is fixed (the AND plane is programmable).
2	**P**rogrammable **L**ogic **D**evice.
3	**PLA** (Programmable Logic Array).
4	2nk + km = 2 × 4 × 6 + 6 × 3 = 48 + 18 = **66**.
5	2^3 words × 2 bits = **16 bits**.

Q	Solution
6	2^4 = **16** rows (one per minterm).
7	2 × 16 × 48 + 48 × 8 = 1536 + 384 = **1920**.
8	PLA: the term A'B is shared → AB', A'B, C = **3** product terms. PAL: F1 needs 2 rows and F2 needs 2 rows (A'B is repeated) → **4** rows.
9	2^4 = **16 bits**: the truth table (16 output values) of any 4-input function.
10	**PLA** can (programmable OR plane with a free choice of terms). A PROM can also, because its OR plane is programmable. A **PAL cannot** (fixed OR plane).
11	2^6 = **64** words (1 bit each).
12	**SRAM-based** devices are volatile. Example: an **FPGA** (it is reloaded from outside at power-up).

Next: Chapter 18 — ADC and DAC (resolution, quantization, sampling, flash, successive approximation, counter-type ADC, R-2R and binary-weighted DAC).

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